

2024



Accelerating Product Engineering Using AWS Solutions with Vitesco Technologies

Learn how Vitesco Technologies modernized its engineering workflows by building a complete cloud environment in under 18 months using AWS.

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1 day

compared with 1 week to render simul

5x more cores

55% savings

on compute costs

Minutes

cloud resou



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Market trends change quickly, especially for emerging technologies such as hybrid and electric vehicles. For German automotive parts manufacturer [Vitesco Technologies](#) (Vitesco), supporting engineering efficiency is critical to speeding up its time to market. To keep up with consumer demand, Vitesco wanted to empower its engineering teams by providing them with on-demand access to high performance computing (HPC) resources and modernizing its product lifecycle management (PLM) software on Amazon Web Services (AWS).

Opportunity | Architecting Scalable, Agile Product Engineering Workloads Using AWS HPC Solutions

Headquartered in Regensburg, Germany, Vitesco aims to power clean mobility by designing and manufacturing automotive parts for electric and hybrid vehicles. To test out new ideas and products, its engineering teams run complex, large-scale simulations and manage the development through PLM software.

Vitesco originally ran its simulations with on-premises resources and clusters. To support a broad product portfolio and its electrified vehicles strategy, Vitesco wanted to run more simulations at faster speeds. The company faced the decision of either procuring more hardware or working around bottlenecks in its engineering workflows.

“In the past, we had clusters with a fixed number of cores for our simulations,” says Theodor Heindl, head of Engineering Applications at Vitesco. “We had a first come, first served principle, which made it difficult to prioritize urgent simulations. The other option was to build a big simulation farm that would be running day and night, whether we needed it or not.”

On spinning off from Continental AG in 2018, Vitesco needed to build infrastructure for over 350 mission-critical applications, including its PLM software, in 18 months. “We were working against a tight time frame,” says Alwin Perras, service owner for 3D mechanical simulation tools and HPC at Vitesco. “We needed a dynamic system that would help us review and control compute costs.” These requirements led the Vitesco team to architect an HPC environment for running its engineering and simulation workloads on AWS. Vitesco developed a proof of concept in 3 days by working alongside [AWS Professional Services](#)—a team that helps businesses achieve their desired business outcomes using AWS.

Solution | Making Product Engineering and Virtual Desktop Resources Available in Minutes Using AWS Services

To quickly get its systems running on AWS, Vitesco migrated its data records using [Amazon FSx for NetApp ONTAP](#), a fully managed shared storage built on NetApp's popular ONTAP file system. Next, Vitesco's HPC team worked on modernizing Vitesco's on-premises clusters with cloud-based and HPC solutions, including [AWS ParallelCluster](#), an open-source cluster management tool that makes it simple for businesses to deploy and manage HPC clusters on AWS. Using cloud-native services, Vitesco can respond better to business needs by automatically scaling its infrastructure up to 4,100 cores. This is five times the number of cores that Vitesco had in its on-premises environment.

To provide centralized data storage to its engineering teams, Vitesco stores its data in [Amazon Simple Storage Service](#) (Amazon S3), an object storage service built to retrieve virtually any amount of data from anywhere. As a result, engineers save precious time because they don't have to download simulation results to their computers. They can also securely share and collaborate on their results from nearly any location. Across the company, Vitesco has stored more than 3 PB of data on AWS. "Our system supports agile development and engineering," says Perras. "Our teams can exchange data and results in one central environment without having to think about it."

In a matter of minutes, Vitesco's engineers can now spin up the resources that they need, such as [Amazon Elastic Compute Cloud](#) (Amazon EC2) [G4 Instances](#) and [Hpc6a Instances](#) for secure and resizable compute capacity for virtually any workload. Vitesco's engineers can choose their instance types based on the urgency of the project and estimated cost. For example, they can run simulations with compute instances, such as the Amazon EC2 Hpc6a Instances, and perform 3D rendering with graphics instances, such as the Amazon EC2 G4dn Instances. In turn, Vitesco has saved 55 percent on compute costs compared with operating on premises. "Our global teams really need speed of development and simulations to deliver business value," says Heindl. "AWS provides huge flexibility and the ability to ramp up our infrastructure and try out new ideas."

Unlike in the on-premises environment, Vitesco's engineers can run simulation jobs simultaneously without causing bottlenecks in their workflows. The runtime of their simulations has also improved. Simulations that used to take 1 week are now delivered in 1 day, helping the company ship new products to customers faster than before. "One of the biggest benefits we've experienced is speed," says Heindl. "Our teams can immediately derive insights from their simulations to further develop our products."

Outcome | Building a Seamless Digital Engineering Environment on AWS

Looking forward, Vitesco wants to move toward developing cloud-based applications and taking advantage of automation potential.

"This transformation has been a big step in accelerating our engineering processes and making them more efficient," says Heindl. "In the future, we're going to keep modernizing our software architecture to be able to use multiple cores instead of single cores. That will speed up our build processes and help our engineers to keep pioneering."

About Vitesco Technologies



Vitesco is a leading developer and manufacturer of modern drive systems for sustainable mobility. With intelligent system solutions for electric, hybrid, and internal combustion drive systems, Vitesco is making mobility clean, efficient, and affordable.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS ParallelCluster

AWS ParallelCluster is an open source cluster management tool that makes it easy for you to deploy and manage High Performance Computing (HPC) clusters on AWS.

[Learn more »](#)

Amazon FSx for NetApp ONTAP

Amazon FSx for NetApp ONTAP provides fully managed shared storage in the AWS Cloud with the popular data access and management capabilities of ONTAP.

[Learn more »](#)

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry scalability, data availability, security, and performance.



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Continental on AWS

From one of the most recognized tire brands in the world to a connected mobility pioneer, Continental has seen success from 1871 to today thanks to its constant evolution. Using Amazon Web Services (AWS), the company has embraced the cloud and accelerated a software-driven vision. It built modular solution Continental Automotive Edge (CAEdge), which connects vehicles to the cloud and provides a virtual



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ITALY



Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best

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possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

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